



COMPUTER SECURITY

2025

FINAL 2024

(Solved)



SENIOR
25

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Choose the correct answer:

1.	A company wants to block the connection between its LAN and “youtube “, simply they can apply _____							
	a)	Packet filtering gateways	b)	Application proxies firewall	c)	Stateful inspection firewalls	d)	Guards firewalls
2.	A company wants to set up an online price list so that outsiders can see the products and prices offered. It wants to be sure that <u>no outsider can change the prices or product list</u> , <i>the outsiders can access only the price list</i> , not any of the more sensitive files stored inside.							
	a)	Packet filtering gateways	b)	Application proxies firewall	c)	Stateful inspection firewalls	d)	Guards firewalls
3.	A company wants to allow its employees to use e-mail up to a limit of so many messages or so many characters of e-mail in the last so many days, simply they can apply _____							
	a)	Packet filtering gateways	b)	Application proxies firewall	c)	Stateful inspection firewalls	d)	Guards firewalls
4.	Which of the following is best description of Stateful inspection firewall working principle?							
	a)	It can see address and services protocol type	b)	It can see full data portion of packet	c)	It can see either address or data	d)	It can see full text of communication
5.	What is true about IDS?							
	a)	it could be stand-alone device attached to the network to monitor traffic throughout that network						
	b)	Honey pot is considered an example for IDS						
	c)	Regarding anomaly-based IDS, admin can mark a flagged behavior as acceptable or not.						
	d)	All						
6.	_____ means intercepting and carrying on a session begun by another entity .							
	a)	Masquerade	b)	Man-in-the-Middle Attack	c)	Session hijacking	d)	Impersonation
7.	Which of the following can describe the function of Firewall system?							
	a)	To encrypt all network traffic to prevent eavesdropping						
	b)	To block all incoming network traffic except for authorized sources						
	c)	To prevent all unauthorized activity of networks’ internal users						
	d)	To monitor internal network traffic and detect potential security breaches						
8.	Denning’s model is an example for _____							
	a)	heuristic based IDS	b)	signature based IDS	c)	Virus Scan software	d)	Firewall model
9.	Which profile is recommended for auditing?							
	a)	logon and session activity	b)	command or program usage	c)	file access activity	d)	All
10.	When OS runs different process of different security requirements on the same time with shared resources under illusion that no other processes exist							
	a)	Cryptographic separation	b)	Temporal separation	c)	logical separation	d)	Physical separation
11.	Physical OS separation approach _____ the <u>security level</u> and _____ the <u>resource sharing</u>							
	a)	Minimizes, minimizes	b)	Maximizes, minimizes	c)	minimizes, maximizes	d)	maximizes, maximizes



12.	if different processes can run concurrently unaware of the presence of each other , and each process has its own address space, files, and other objects. This OS level of sharing is categorized as:							
	a)	Isolation level	b)	Share via capabilities	c)	Share via access limitations	d)	Share all or share nothing
13.	Triple-DES with two keys is defined by $C = Ek_1(Dk_2(Ek_1(P)))$, if K1 equals K2 , the triple-DE will be:							
	a)	Double DES	b)	Triple-DES	c)	Single-DES	d)	Useless encryption
14.	What is true about Kerberos authentication protocol?							
	a)	It is a two-party, two-way authentication protocol						
	b)	It is a two-party, three-way authentication protocol						
	c)	It is a three-party, two-way authentication protocol						
	d)	It is a three-party, three-way authentication protocol						
15.	The ticket issued to the user by Kerberos protocol includes the following data							
	a)	User login details	b)	Time-date stamp	c)	User privilege on the accessed application	d)	All are true
16.	What is true about Steganography?							
	a)	it is used to encrypt images and send it in unreadable forms						
	b)	The encrypted file has smaller size compared with the original one						
	c)	it can be broken by brute-force attack						
	d)	it uses images to hide important data in LSB of each image's pixel						
17.	Choose the correct sentence?							
	a)	Erasure of a program is considered an interruption threat						
	b)	TOCTOU is considered an attack of availability						
	c)	Flooding are attacks of confidentiality						
	d)	Error is a mistake in interpreting a syntax error in a piece of code						
18.	What is a common example of a man-in-the-middle attack							
	a)	Trojan horse	b)	DDoS attack	c)	Wi-Fi eavesdropping	d)	SQL injection
19.	How can a man-in-the-middle attack be prevented?							
	a)	Using strong encryption	b)	Using secure authentication methods	c)	Using digital signatures	d)	All of the above
20.	How can a user protect themselves from a phishing attack?							
	a)	Being cautious of messages or websites that ask for sensitive information						
	b)	Checking the URL of a website before entering any information						
	c)	Try to depend on two way -two party authentication techniques						
	d)	All of the above						



21.	Based on the Playfair matrix of Fig. 1, the plaintext of the cipher text "NXOMTW" is _____ NX → PV OM → UH TW → AO Plaintext: PVUHAO				M	F	H	I/J	K
					U	N	O	P	Q
					Z	V	W	X	Y
					E	L	A	R	G
					D	S	T	B	C
					M	F	H	I/J	K
					U	N	O	P	Q
					Z	V	W	X	Y
					E	L	A	R	G
					D	S	T	B	C
a) Hellox		b) Sunset		c) Party		d) None			

22.	_____ means that different processes use different physical objects , such as separate printers for output requiring different levels of security						
a)	Temporal separation	b)	Physical separation	c)	Logical separation	d)	Cryptographic separation

23.	A type of virus which is active as long as the computer running						
a)	Memory resident virus	b)	Brain virus	c)	Polymorphic virus	d)	Logic bomb

24.	controls against program threats include the following actions except_____						
a)	peer review process	b)	Independent testing	c)	Proof of correctness	d)	Decapsulation of different process

25.	Each virus has a signature that could be used to detect its existence, such as:						
a)	Its location	b)	its execution performance	c)	its replication technique	d)	All

26.	Viruses resident in application programs and libraries are categorized as:						
a)	One time execution virus	b)	Boot sector virus	c)	Other homes of virus	d)	Memory resident

27.	A program virus attaches itself to a program ; then, whenever the program is run, the virus is activated. This kind of attachment is usually known as _____						
a)	integrated virus	b)	Surrounded virus	c)	Appended virus	d)	replacement virus

28.	What is a weakness of Caesar cipher?						
a)	Vulnerable to brute-force attacks	b)	Vulnerable to frequency analysis	c)	limited key space	d)	all

29.	What is the number of rounds in <u>single DES</u> ?						
a)	64	b)	56	c)	48	d)	16

30.	Encrypting P= "COMPUTER NETWORK" by Block transposition algorithm k=32514, then C is: COMPU TERNE TWORK MOUCP REETN OWKTR						
a)	PNROEWCTTUEKMRO	b)	TWORKTERNECOMPU	c)	CTTOEWMROPNRUEK	d)	None



True/False:

1.	The firewall can't be able to detect the signature of the attack if it was broken into multiple packets each have very short length	T
2.	A personal firewall is an application program that runs on a user's workstation usually under his policy	T
3.	Auditing is difficult when applying Packet filtering firewalls	T
4.	Heuristic based IDS performs simple pattern-matching to report a known attack type	F
5.	<u>Verification of the user's identity</u> is known as authorization process (authentication)	F
6.	Digital signatures provide the ability to verify both owner and content of message	T
7.	The plain text must be broken into a block size of 128bit as an input of <u>Triple DES</u> (64-bit)	F
8.	Polymorphic virus is a form of virus explicitly designed to hide itself from detection	T
9.	A boot sector virus changes control on the OS instructions, when a computer is started	T
10.	A network-based IDS is installed on the network while a host-based IDS is installed on individual hosts.	T
11.	Playfair cipher rearranges the order of words in a message (letters)	F
12.	Shift cipher applies a <u>mathematical formula</u> to each letter in a message that replaces each letter with a numerical value. (Vigenère)	F
13.	Flooding is an attack of availability	T
14.	Authorization is mechanism used to confirm a user's identity (authentication)	F
15.	Fingerprint is used to verify the user as an example for something user has	F
16.	Cryptographic separation is applied when OS runs different process of different security requirements on different time with separate resources (temporal)	F
17.	Trapdoor is a malicious software runs when a computer is started, it Changes control on the OS instructions (Boot sector virus)	F
18.	Program controls must include peer review steps	T
19.	In context of Program controls, proof of program correctness is done based on check list	T
20.	Code flaws include validation error or domain error	T

